

Pranshu Upadhyay

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Software Engineer — AI Systems & Full-Stack

EDUCATION

B.Tech, Computer Science & Engineering (AI & Robotics) — VIT Chennai

CGPA: 8.69

BS, Data Science and Programming — IIT Madras (concurrent)

Expected 2028

PROJECTS

Connected Vehicle Telemetry Dashboard — FastAPI, WebSockets, React, PostgreSQL, Redis, Docker

github.com/upadhyayPranshu/fleet-telemetry-dashboard

- Built a real-time fleet telemetry system simulating 200+ vehicles, streaming live GPS, fuel, battery, engine temperature, RPM, speed, and tire pressure over WebSockets
- Designed a Redis-backed live-state cache and pub/sub fanout so the dashboard never blocks on a database read, with PostgreSQL handling batched historical writes separately
- Implemented a weighted health-scoring model (worst-factor-weighted, not a flat average) and threshold-based alerting for abnormal readings
- Containerized the full stack (FastAPI, React, PostgreSQL, Redis, Nginx) with Docker Compose for one-command deployment

AI RFQ Verifier — React 19, FastAPI, MongoDB, Ollama, SentenceTransformers

github.com/upadhyayPranshu/rfq_verifier

- Architected a 3-layer AI evaluation pipeline (regex pre-check → LLM fact extraction → LLM compliance judgment) to verify vendor bids against RFQ clauses
- Implemented hybrid semantic search with page-level bi-encoder embeddings to route clauses to the most relevant bid pages before invoking the LLM, avoiding unnecessary model calls
- Built the full stack end to end: React 19 + Vite frontend, FastAPI backend, MongoDB, local LLM inference via Ollama to avoid external API costs and data exposure
- Combined pdfplumber, PyMuPDF, PyPDF2 and Tesseract OCR to handle both text-based and scanned bid PDFs

Letter Rush — React Native, TypeScript, Firebase

github.com/upadhyayPranshu/vocab_antakshari

- Built a cross-platform multiplayer vocabulary game with real-time and offline modes, using Firebase for room-based multiplayer sync
- Implemented turn-based logic, timers, and a light/dark theme system

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript

Frontend: React, React Native, Vite

Backend: FastAPI, Node.js, REST APIs, WebSockets

AI / ML: LLM pipeline design, Ollama, SentenceTransformers, hybrid semantic search

Databases: PostgreSQL, MongoDB, Redis, Firebase

DevOps / Cloud: Docker, Docker Compose, Firebase

Dev Tools: Git, GitHub, ESLint, Prettier, Jest

CODING PROFILE

150+ problems solved on LeetCode (C, C++, Java) — advanced-tier strength in Dynamic Programming, Backtracking, and Divide & Conquer; consistent practice reflected in a 50-Day Streak Badge (2025).